26/7/26

# Exp - 2 Experiment On Packet Capture Tool

Aim: To understand the feature of wireshark as a Packet capture tool and understand encapsulation of information at various layers of a protocol stack

Packet Sniffer :-

* Capture Packet sent and received by the computer
* Display protocol information of captured Packet
* Work Passively
* Retain copies of network Packet for analysis

Packet Sniffer tool :-

* Tcp dump - Command line Packet capture tool
* Wireshark - Graphical Packet analyzer

Description:

Wireshark is a network Protocol analyzer that captures and displays network Packet in real time. It helps monitor, analyze, troubleshoot & inspect network traffic using filters and Packet details.



## Uses of Network

(i) Capture network traffic
(ii) Analyze protocol
(iii) Apply display filters
(iv) Monitor network issue

## Application

(i) Network troubleshooting
(ii) Security analysis
(iii) Protocol debugging
(iv) Learning network protocol

## Capturing Packet:

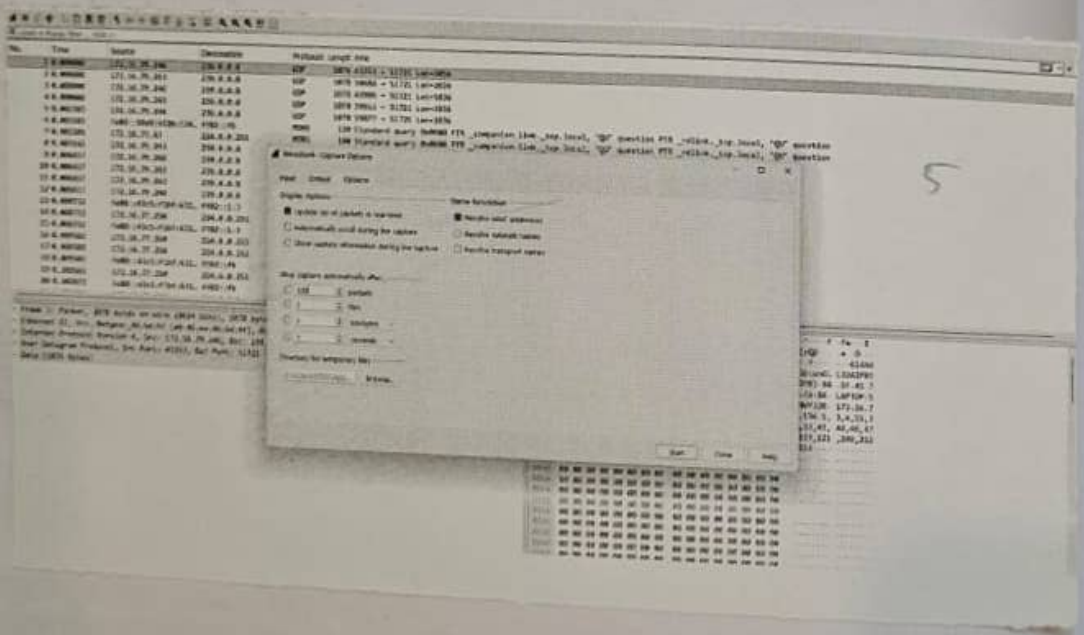
(i) Open Wireshark
(ii) Select the network interface
(iii) Click start capture
(iv) Observe Packet in real time
(v) Click stop and save to the capture

## Wireshark Pane:

(i) Packet list: Display capture Packets
(ii) Packet details: Shows Protocol information
(iii) Packet Bytes: Display Packet data in Hexadecimal Point

## Color Coding:-

(i) Purple :- TCP
(ii) Blue : UDP
(iii) Black : Error Packet.



Filtering Packets:

* Enter a filter in the filter bar
* Right click to apply filter (or) follows the Top stream

Flow graph:

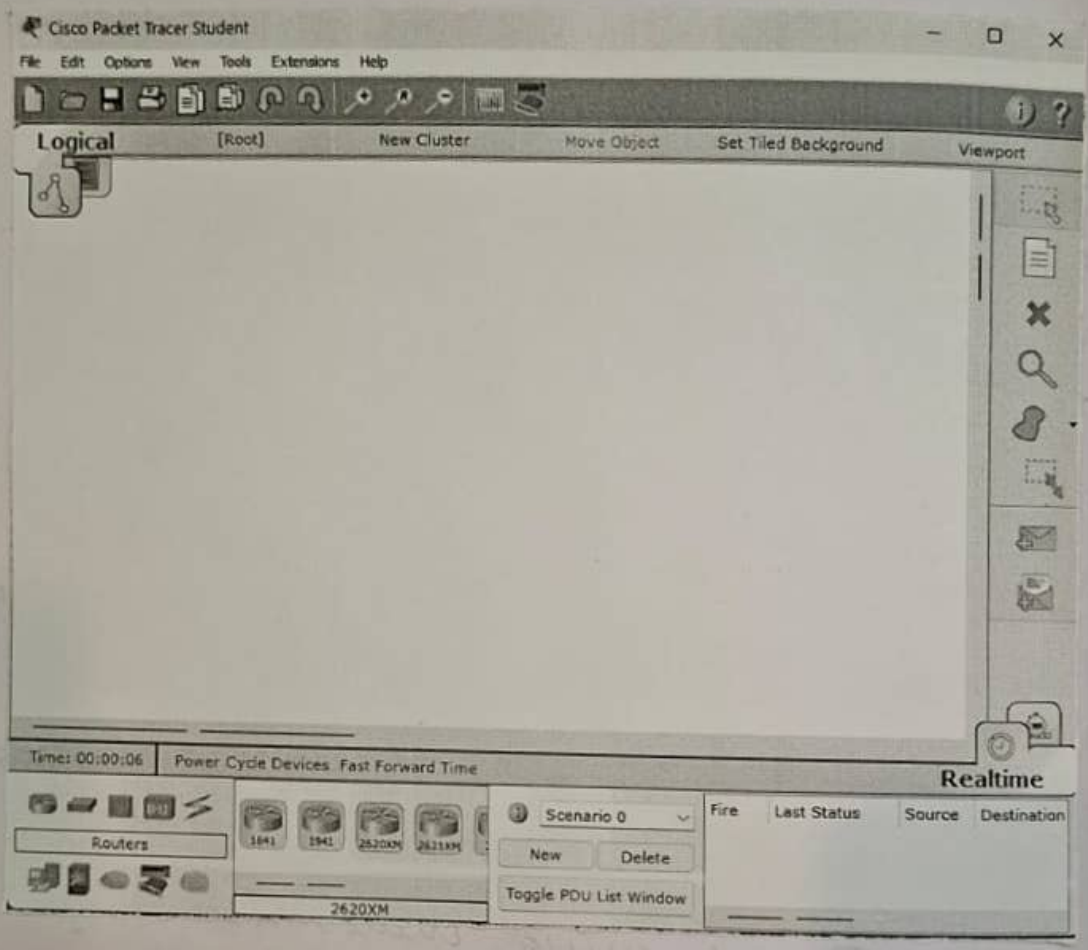
Displays the communication flow b/w device, making package, exchange. easier to understand

Capture 100 Packets

① Open wireshark
② Select the network interface.
③ Set the capture limit into 100 Packets
④ Start Capturing
⑤ Stop and save the captured Packets

Analyzing Packet using wireshark Tools:

① Create a filter to display only TCP/UDP Packets
   ① Open the capture file
   ② Apply the TCP || UDP filter
   ③ Inspect and View the flow graph
② Creat a file to display only ARP Packets.
   ① Open the capture file
   ② Apply the ARP filter
   ③ Inspect and View the flow graph.


Cisco Packet Tracer Student
File Edit Options View Tools Extensions Help
Logical
[Root]
New Cluster
Move Object
Set Tiled Background
Viewport
Time: 00:00:06
Power Cycle Devices Fast Forward Time
Realtime
Routers
2620XM
Scenario 0
New
Delete
Toggle PDU List Window
Fire
Last Status
Source
Destination

3. Create a filter to display only DNS Packet

(i) Open the capture file

(ii) Apply the DNS filter

(iii) Inspect and view the flow graph

(4) Create a filter to display only IP/Jamp

(i) Open the capture file

(ii) Apply the ip or tcmp filter.

(iii) Inspect and view the flow graph and Save capture.

(5) Create a filter to display only DHCP Packet

(i) Open the capture file

(ii) Apply the dhcp filter

(iii) Inspect and view the flow graph and Save the capture.

Result:

Thus the feature of wireshark and the implement of different filter was successfully compiled.